

# Exercise-Induced Fatigue in Hamstring versus Quadriceps Muscles and Consequences on the Torque–Duration Relationship in Men

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<sup>1</sup>Faculty of Medicine, Mitochondria, Oxidative Stress and Muscular Protection laboratory (UR 3072), University of Strasbourg, Strasbourg, FRANCE; <sup>2</sup>Faculty of Sport Sciences, European Centre for Education, Research and Innovation in Exercise Physiology (CEERIPE), University of Strasbourg, Strasbourg, FRANCE; and <sup>3</sup>Racing Club de Strasbourg Alsace, Strasbourg, FRANCE

## ABSTRACT

MASSAMBA, A., E. HUCTEAU, J. MALLARD, G. P. DUCROCQ, F. FAVRET, and T. J. HUREAU. Exercise-Induced Fatigue in Hamstring versus Quadriceps Muscles and Consequences on the Torque–Duration Relationship in Men. *Med. Sci. Sports Exerc.*, Vol. 54, No. 12, pp. 2099–2108, 2022. **Purpose:** The present study investigated the mechanisms of neuromuscular fatigue in quadriceps and hamstring muscles and its consequences on the torque–duration relationship. **Methods:** Twelve healthy men performed a 5-min all-out exercise (3-s contraction, 2-s relaxation) with either quadriceps or hamstring muscles on separate days. Central fatigue and peripheral fatigue were quantified via changes in pre- to postexercise voluntary activation (VA) and potentiated twitch ( $P_{TW}$ ) torque evoked by supramaximal electrical stimulation, respectively. Critical torque was determined as the mean torque of the last six contractions, whereas  $W'$  was calculated as the torque impulse done above critical torque. **Results:** After exercise, maximal voluntary contraction (MVC) decreased to a greater magnitude ( $P < 0.001$ ) in quadriceps ( $-67\% \pm 9\%$ ) compared with hamstring ( $-51\% \pm 10\%$ ).  $\Delta P_{TW}$  was also greater in quadriceps compared with hamstring ( $-69\% \pm 15\%$  vs  $55\% \pm 10\%$ ,  $P < 0.01$ ), whereas central fatigue only developed in quadriceps ( $\Delta VA$ ,  $-25\% \pm 28\%$ ). Hamstring demonstrated reduced critical torque compared with quadriceps ( $60 \pm 12$  vs  $97 \pm 26$  N·m,  $P < 0.001$ ) as well as drastically lower  $W'$  ( $1001 \pm 696$  vs  $8111 \pm 2073$  N·m·s,  $P < 0.001$ ). No correlation was found between quadriceps and hamstring for any index of neuromuscular fatigue ( $\Delta MVC$ ,  $\Delta P_{TW}$ , or  $\Delta VA$ ). **Conclusions:** These findings revealed that hamstring presented different etiology and magnitude of neuromuscular fatigue compared with quadriceps. The absence of correlation observed between quadriceps and hamstring fatigue parameters ( $\Delta MVC$ ,  $\Delta P_{TW}$ , or  $\Delta VA$ ) suggests no interrelation in fatigue etiology between these two muscle groups within individuals and, therefore, highlights the need to investigate specifically hamstring muscle fatigue. **Key Words:** CENTRAL AND PERIPHERAL FATIGUE, HAMSTRING MUSCLE, CRITICAL TORQUE, EXERCISE PERFORMANCE, RECOVERY

The hamstring muscles play a major role in sprinting performance (1) and are a common location for severe muscle injuries (2). Exercise-induced neuromuscular fatigue has been suggested as a risk factor for hamstring muscle injuries, as indirectly evidenced by a greater injury incidence in the later stages of soccer (2) or rugby (3) matches. Neuromuscular fatigue is attributed to reductions in maximal voluntary contraction (MVC) because of central and/or peripheral factors. Central fatigue refers to a failure of the central nervous system to maintain force, resulting in a reduction in

descending neural drive and/or output from the spinal motoneurons (4). Peripheral fatigue is the result of biochemical changes within skeletal muscle, yielding an attenuated response to neural stimulation (5). Central and peripheral fatigue can be quantified noninvasively via reductions in the voluntary muscle activation and in the resting twitch force, respectively.

The proportion of fatigue-resistant Type I muscle fibers is greater in hamstring compared with quadriceps muscles (6,7), suggesting that the hamstring muscles are less fatigable than the quadriceps muscles. However, Coratella et al. (8) demonstrated greater neuromuscular fatigue in hamstring than quadriceps after intermittent isometric exercise performed at the same relative intensity (70% of the MVC). Nevertheless, in this scenario, time to task failure was drastically different between muscle groups, with quadriceps exercising two times longer than hamstring before reaching task failure. Importantly, it has been documented that exercise duration/intensity is a strong determinant of neuromuscular fatigue etiology (9), with relatively more peripheral fatigue, and less central fatigue, for short/intense trials compared with longer/less intense trials. To circumvent the confounding effect associated with differences in exercise

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duration/intensity while comparing quadriceps and hamstring muscle groups, the use of an intermittent all-out exercise protocol might be insightful (10,11).

Moreover, the use of an intermittent all-out exercise presents the opportunity to assess the development of neuromuscular fatigue throughout exercise (i.e., using rests between contractions). In this context, key parameters derived from the torque–duration relationship provide a cohesive framework within which to investigate the mechanistic bases of neuromuscular fatigue and exercise performance. Critical torque is mathematically defined as the asymptote of the hyperbolic torque–duration relationship, whereas the curvature constant ( $W'$ ) represents the fixed amount of torque impulse that can be performed above critical torque (12). Physiologically, critical torque is associated with the greatest oxidative metabolic rate that can be sustained without a continuous reduction in  $W'$  (13). Vanhatalo et al. (13) observed a positive correlation between critical power (equivalent to critical torque during a cycling exercise) and the proportion of fatigue-resistant Type I muscle fibers. Moreover, the magnitude of  $W'$  depleted was associated with the development of both neuromuscular fatigue (14–17) and maximal force capacity (18). Therefore, the characterization of critical torque and  $W'$  would provide insights into exercise performance and neuromuscular fatigue for the comparison of hamstring and quadriceps muscles.

The present study aimed to investigate the mechanisms of hamstring neuromuscular fatigue and recovery, compared with quadriceps, during all-out exercise to match both exercise duration and relative intensity between muscle groups. Moreover, we sought to determine and compare the parameters of the torque–duration relationship between hamstring and quadriceps. We hypothesized that hamstring would develop less neuromuscular fatigue than quadriceps muscles. As a consequence of this presumably lower fatigability, recovery from fatiguing exercise would be slower, and  $W'$  and critical torque (in absolute) would be lower, in hamstring compared with quadriceps.

## METHODS

**Subjects.** Twelve healthy men participated in the present study (mean  $\pm$  SD: age,  $26 \pm 4$  yr; height,  $174 \pm 7$  cm; body mass,  $76 \pm 17$  kg; body fat percentage,  $14.8\% \pm 5.6\%$ ). Body mass and composition were assessed by using a bioelectrical impedance meter (Tanita DC-13C; Tanita Corporation, Tokyo, Japan). All subjects were recreational athletes, nonsmokers, and were not taking any medications. They were requested to prohibit physical activity for 24 h before each experimental session. Written informed consent was obtained from each participant before the beginning of the study. The study was approved by the local ethics committee and conducted according to the Declaration of Helsinki for human experimentation.

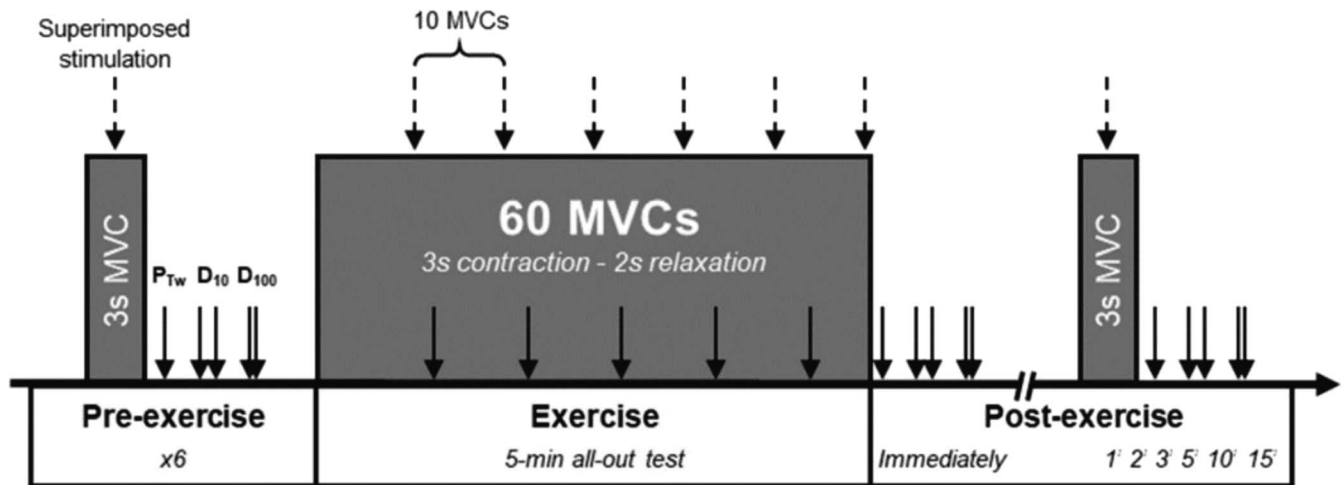
**Experimental design.** After familiarization sessions (one for each muscle group), all participants visited the laboratory for two experimental sessions where they performed a fatiguing exercise protocol with either quadriceps or hamstring

muscles in a randomized order. Each visit was separated by at least 48 h and performed with the right leg at the same time of day ( $\pm 1$  h).

**Experimental protocol.** Sessions started with a standardized warm-up consisting of a set of 10 graded submaximal isometric contractions with visual feedback displayed on a computer screen. All torque measurements were conducted using an isokinetic dynamometer (Biodex System 3; Biodex Medical Systems, Inc., Shirley, NY), following standardized procedures and instructions given by the same examiner. Participants were seated on the dynamometer with  $85^\circ$  of hip flexion for quadriceps measurements while they laid prone with  $10^\circ$  of hip flexion for hamstring measurements. They were stabilized with safety belts strapped across the chest and hips to avoid lateral, vertical, or frontal displacements. Torque (N·m) was collected from a lever arm attached to the subject by a non-compliant strap located  $\sim 2$  cm superior to the lateral malleolus. To ensure maximal torque development, each quadriceps and hamstring measurement was conducted at an optimal angle, namely, the angle where torque output was maximal (i.e.,  $83^\circ \pm 6^\circ$  and  $28^\circ \pm 8^\circ$ , respectively). The isometric torque–angle relationship was determined on an individual basis during the familiarization sessions. To this end, participants performed 3 MVC interspersed by 1 min of rest at each angle tested (every  $10^\circ$  from  $0^\circ$  to  $90^\circ$  in hamstring and from  $50^\circ$  to  $90^\circ$  in quadriceps, where  $0^\circ$  is full knee extension). If the coefficient of variation between the three MVC was greater than 5%, a fourth MVC was completed. Thereafter, neuromuscular function assessments and exercise procedures (i.e., 60 MVC) were conducted.

Participants performed 60 intermittent isometric single-leg MVC over a 5-min period (3-s contraction/2-s relaxation) with either quadriceps or hamstring muscles. A recording cued the start and stop of each 3-s contraction. A visual feedback of torque output was provided on the computer screen, and participants were strongly encouraged to maximize performance. To ensure that subjects reached their best performance during the first contraction, all participants had to satisfy the criterion score corresponding to 95% of their best MVC performed during baseline measurement. The development of neuromuscular fatigue was tracked during the 5-min all-out test by delivering superimposed and potentiated twitches every 10 contractions (Fig. 1).

**Neuromuscular function assessment.** Neuromuscular tests were conducted before, during (every 10 MVC), and directly after (60th MVC) the fatiguing exercise protocol. Recovery was assessed at 1, 2, 3, 5, 10, and 15 min postexercise. A constant current stimulator (DS7AH; Digitimer Limited, Hertfordshire, UK) delivered single square-wave stimuli of 0.2-ms duration with maximal voltage of 400 V over the femoral nerve (quadriceps), or over the muscle belly (hamstring). For quadriceps, the cathode (self-adhesive electrode:  $3 \times 3$  cm, Ag/AgCl, Mini-KR; Contrôle-Graphique, Brie-Comte-Robert, France) was positioned on the femoral nerve. The anode, a  $10 \times 5$  cm self-adhesive stimulation electrode (Compex Medical SA, Ecublens, Switzerland), was placed midway between



**FIGURE 1**—Schematic diagram of the experimental protocol. After familiarization sessions (one for each muscle group), all participants visited the laboratory for two experimental sessions where they performed a fatiguing 5-min all-out exercise protocol (3-s contraction, 2-s relaxation) with either quadriceps or hamstring muscles. After a standardized warm-up, the neuromuscular function was assessed preexercise to determine baseline values that were used to quantify neuromuscular fatigue development during exercise (assessment every 10 MVC) and short-term recovery (postexercise assessments). Central fatigue and peripheral fatigue were quantified via changes in pre- to postexercise VA and  $P_{TW}$  torque (single twitch,  $P_{TW}$ ; doublets at 10 Hz,  $D_{10}$ ; doublets at 100 Hz,  $D_{100}$ ) evoked by supramaximal electrical stimulation, respectively. Dashed arrows depict superimposed stimulations during the MVC, and solid arrows indicate potentiated stimulations delivered at rest.

the greater trochanter and the iliac crest. Direct muscle stimulation was used for hamstring using two self-adhesive electrodes ( $10 \times 5$  cm, Compex Medical SA), with the cathode placed immediately below the gluteal fold and the anode approximately 5 cm superior to the popliteal fossa (19). Electrode positions were marked with indelible ink to ensure a reproducible stimulation between visits. To ensure supramaximality during these tests, the stimulation intensity (quadriceps,  $136 \pm 54$  mA; hamstring,  $176 \pm 47$  mA) was set to 120% of the intensity eliciting the maximal spatial recruitment of the targeted muscle, as confirmed by both maximal torque response and maximal vastus lateralis or biceps femoral muscle compound action potential ( $M_{max}$ ) for quadriceps and hamstring, respectively. Participants subsequently completed six MVC (3-s duration) before exercise, separated by 30 s rest. For the final four contractions, single potentiated twitch ( $P_{TW}$ ), 10 Hz doublet ( $D_{10}$ ), and 100 Hz doublet ( $D_{100}$ ) were delivered 1, 3, and 5 s after every MVC, respectively. For all MVC,  $D_{10}$ , and  $D_{100}$ , peak torque was determined. In addition, the following mechanical measures of muscle contractility were derived from  $P_{TW}$ : peak torque, contraction time, maximal rate of torque development (MRTD), and half relaxation time. During MVC, superimposed  $P_{TW}$  were delivered to estimate voluntary activation (VA) according to the following formula (20):

$$VA (\%) = [1 - (\text{superimposed twitch}) / (P_{TW})] \times 100.$$

Peripheral and central fatigue levels were quantified via changes in preexercise to postexercise  $P_{TW}$  torque and VA, respectively. Global (peripheral + central) fatigue was determined as the difference in MVC from baseline.

**Surface EMG.** Electrical activities of quadriceps and hamstring were assessed using surface EMG (Octal Bio-Amp.; ADInstruments, Bella Vista, NSW, Australia). Bipolar pregelled

Ag/AgCL electrodes (15-mm interelectrode distance) were placed over biceps femoris (BF), and semitendinosus (ST) for hamstring, as well as over vastus lateralis (VL), vastus medialis (VM), and rectus femoris (RF) for quadriceps. For each visit, the skin was shaved, abraded with emery paper, and cleaned with alcohol. Electrodes were placed according to the SENIAM recommendations (21). Electrode positions had to be adjusted to optimize M-wave shape and amplitude. The reference electrode was placed over the head of the fibula. The EMG signals were amplified, filtered with a bandwidth frequency ranging from 10 Hz to 1 kHz and recorded at a sampling frequency of 4 kHz using a commercially software (LabChart 8). During the 5-min all-out exercise, muscle activation was calculated for each muscle group with the root-mean-square (RMS) on EMG signals. Specifically, the RMS was determined over a fixed 500-ms window starting 250 ms before peak torque. For data analysis, the muscle activation was normalized by the maximal amplitude of the M-wave ( $RMS \cdot M_{max}^{-1}$ ).

**Determination of critical torque and  $W'$ .** The critical torque was estimated as the mean torque recorded over the last six contractions of the 5-min all-out test, whereas  $W'$  was calculated as the total torque impulse ( $N \cdot m \cdot s$ ) done above critical torque (10).

**Statistics.** Descriptive statistics are presented in the Results section as mean  $\pm$  SD. Normality of all dependent variables and sphericity of variance of the distribution were assessed using the Kolmogorov–Smirnov test and the Mauchly test, respectively. A Greenhouse–Geisser correction was used when sphericity was violated. A two-way (condition–time) repeated-measures ANOVA was used to assess differences in neuromuscular fatigue indices ( $\Delta MVC$ ,  $\Delta P_{TW}$ , VA, and  $RMS \cdot M_{max}^{-1}$ ) between muscle groups (hamstring vs quadriceps) and over time during exercise period. A second two-way (condition–time) repeated-measures ANOVA was performed to

compare differences between conditions (hamstring vs quadriceps) and over time for the recovery period (Pre, Post, and 1, 2, 3, 5, 10, and 15 min postexercise). When a significant difference was identified, multiple-comparison analysis was performed using Tukey *post hoc* test. Student's paired *t*-test was performed for variables relative to torque–duration relationship (i.e., critical torque and  $W'$ ). Relationship between MVC and  $\Delta$ MVC postexercise and fatigue parameters between muscle groups were tested by calculating individual Person correlation coefficients ( $r^2$ ). Statistical analyses were conducted using GraphPad Prism, v8.3.0 (GraphPad, La Jolla, CA). Statistical significance was set at  $P < 0.05$ .

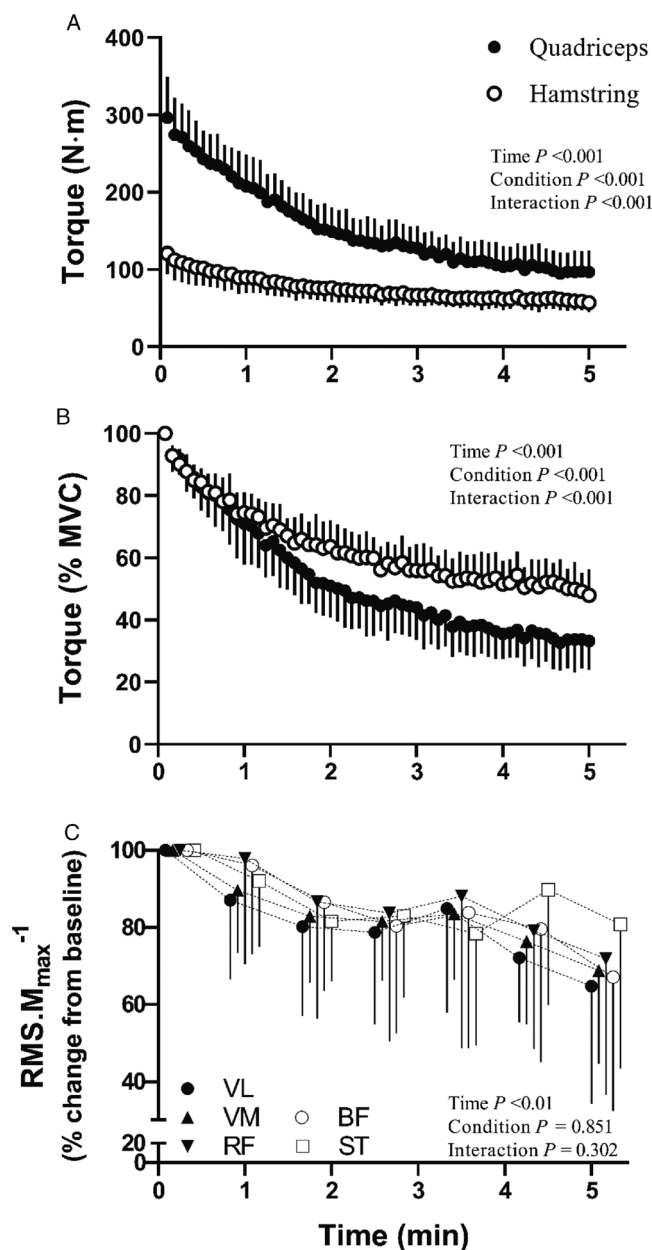
## RESULTS

**Baseline.** Preexercise MVC values did not differ from the first MVC of the 5-min all-out protocol for each muscle group (quadriceps:  $295 \pm 50$  vs  $296 \pm 53$  N·m,  $P = 0.70$ ; hamstring:  $119 \pm 26$  vs  $121 \pm 27$  N·m;  $P = 0.35$ ). As shown in Figure 2A, the torque developed during the first MVC was significantly lower in hamstring than quadriceps ( $P < 0.001$ ).

**Torque output during the 5-min all-out protocol and recovery.** As illustrated in Figure 2, torque output decreased during exercise on average from the first to the 39th MVC in quadriceps and from the first to the 34th MVC in hamstring. Thereafter, torque output plateaued until the last MVC for both muscle groups. The decrease in MVC induced by the all-out exercise protocol ( $\Delta$ MVC) was lower in hamstring ( $-51\% \pm 10\%$ ) compared with quadriceps ( $-67\% \pm 9\%$ ;  $P < 0.001$ ). Interestingly,  $\Delta$ MVC (expressed in newton-meter) was correlated with preexercise MVC ( $r^2 = 0.993$ ,  $P < 0.001$ ), and this relationship was independent of the muscle group (Fig. 3). All muscles (VL, VM, RF, BF, and ST) demonstrated a decline in  $\text{RMS} \cdot \text{M}_{\text{max}}^{-1}$  during exercise ( $P = 0.002$ ), with no difference between hamstring and quadriceps (Fig. 2C).

During the 15-min recovery period (Fig. 4A), MVC did not return to preexercise values ( $-15\% \pm 11\%$  in quadriceps and  $-11\% \pm 9\%$  in hamstring,  $P < 0.001$  vs preexercise). Both muscle groups followed similar recovery kinetics, characterized by a large recovery rate during the first minute postexercise ( $28\% \pm 11\% \cdot \text{min}^{-1}$  in quadriceps vs  $26\% \pm 14\% \cdot \text{min}^{-1}$  in hamstring), followed by a drastically slower recovery rate from 1 to 15 min postexercise ( $1.7\% \pm 0.8\% \cdot \text{min}^{-1}$  in quadriceps vs  $2.9\% \pm 0.7\% \cdot \text{min}^{-1}$  in hamstring).

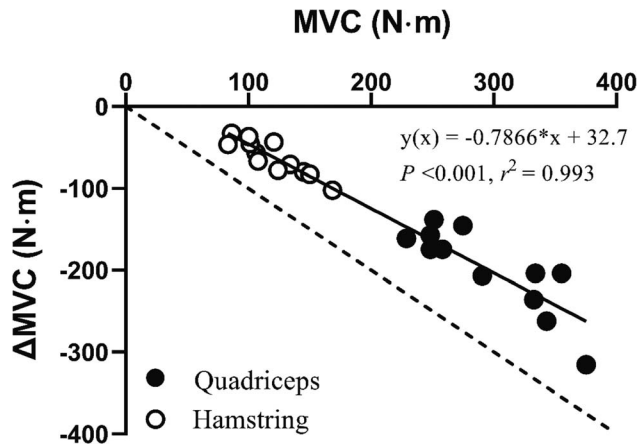
**Peripheral fatigue and recovery ( $\Delta P_{\text{Tw}}$ ,  $D_{10}$ ,  $D_{100}$ , and M-waves).** Neuromuscular fatigue parameters are shown in Figure 4 and Table 1. At the peripheral level, significant interaction effect (i.e., time–muscle group) was found for  $P_{\text{Tw}}$  ( $P < 0.001$ ; Fig. 4B) and  $D_{10}/D_{100}$  ( $P < 0.001$ ; Fig. 4D). The magnitude of peripheral fatigue was lower in hamstring compared with quadriceps ( $\Delta P_{\text{Tw}}$ ,  $-55\% \pm 10\%$  vs  $-69\% \pm 15\%$ ). For both muscle groups, the development of  $\Delta P_{\text{Tw}}$  was large during the first half of exercise (hamstring,  $-20.0\% \pm 2.8\% \cdot \text{min}^{-1}$ ; quadriceps,  $-26.9\% \pm 4.4\% \cdot \text{min}^{-1}$ ), followed by a drastically slower rate during the second half (hamstring,  $-1.9\% \pm 2.6\% \cdot \text{min}^{-1}$ ; quadriceps,  $-0.7\% \pm 3.3\% \cdot \text{min}^{-1}$ ). The decrease in  $D_{10}/D_{100}$ ,



**FIGURE 2—Torque production and muscle activation during the 5-min all-out exercise protocol.** Data are expressed in newton-meter (A) and normalized as a percent of baseline (B) to compare quadriceps and hamstring muscles. Muscle activation (C) was normalized to  $M_{\text{max}}$  for each muscle group.  $M_{\text{max}}$ , maximal M-wave amplitude; RF, rectus femoris; ST, semitendinosus; VL, vastus lateralis; VM, vastus medialis. Data are presented as means  $\pm$  SD. Significant time, condition, and interaction effects are indicated on the graphs.

indicating low-frequency fatigue, was greater in quadriceps compared with hamstring ( $P < 0.001$ ; Fig. 4D). No change in M-waves amplitudes, demonstrating no alteration in membrane excitability after the all-out exercise, was found for all muscles (all  $P > 0.05$ ; Table 1).

Both muscle groups displayed large recovery during the first minute postexercise ( $P_{\text{Tw}}$ ,  $21\% \pm 8\% \cdot \text{min}^{-1}$  in quadriceps vs  $9\% \pm 10\% \cdot \text{min}^{-1}$  in hamstring) followed by a slower recovery rate from 1 to 15 min postexercise ( $P_{\text{Tw}}$ ,  $0.3\% \pm 1.0\% \cdot \text{min}^{-1}$



**FIGURE 3**—Correlation between peak torque and muscle fatigue. MVC peak torque (N·m) was the MVC measured preexercise, whereas muscle fatigue was the decrease in MVC reached from pre- to immediately postexercise ( $\Delta$ MVC, expressed in N·m) in quadriceps and hamstring muscles.

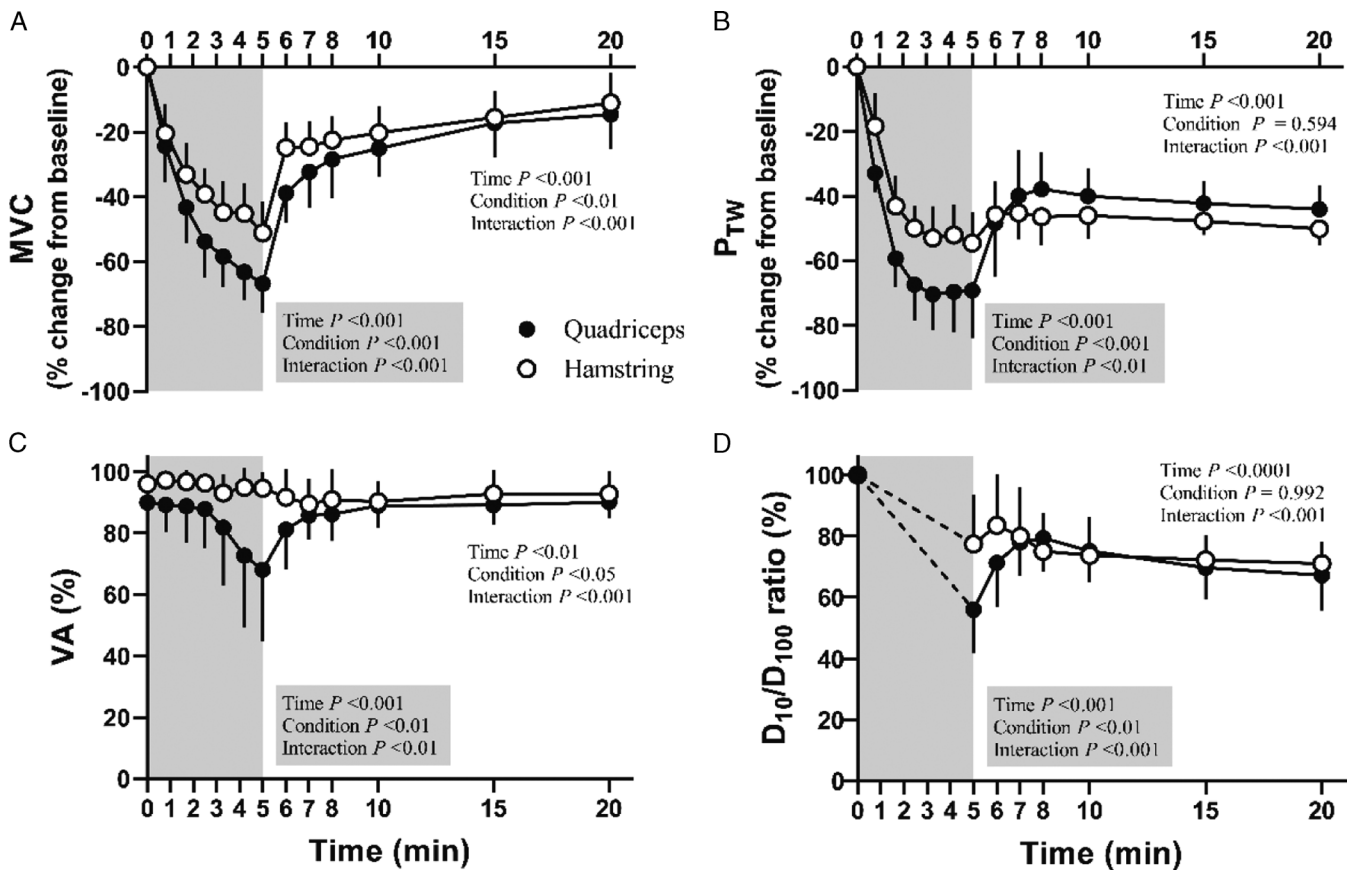
in quadriceps vs  $0.3\% \pm 0.7\% \cdot \text{min}^{-1}$  in hamstring). However, a significant interaction effect demonstrated a slower recovery rate in hamstring compared with quadriceps for  $P_{TW}$  ( $40\% \pm 10\%$  in hamstring vs  $52\% \pm 12\%$  in quadriceps,  $P < 0.001$ ),  $D_{10}$

( $0\% \pm 9\%$  in hamstring vs  $19\% \pm 9\%$  in quadriceps,  $P < 0.001$ ), and  $D_{100}$  ( $4\% \pm 10\%$  in hamstring vs  $17\% \pm 11\%$  in quadriceps,  $P = 0.013$ ). Specifically,  $P_{TW}$  recovery was  $\sim 2$  times slower in hamstring compared with quadriceps during the first minute postexercise. At 15 min postexercise, indicators of peripheral fatigue ( $P_{TW}$ ,  $D_{10}$ ,  $D_{100}$ , and  $D_{10}/D_{100}$ ) did not return to baseline values ( $P < 0.001$ ) in both conditions.

**Central fatigue and recovery (VA).** For central fatigue, a significant interaction effect was found for VA ( $P < 0.001$ ; Fig. 4C). Specifically, although VA was reduced to  $68\% \pm 23\%$  at exercise cessation in quadriceps, it remained unchanged from baseline in hamstring ( $95\% \pm 5\%$ ).

At 15 min postexercise, VA returned to baseline ( $P = 0.999$  for quadriceps and  $P = 0.941$  for hamstring). As illustrated in Figure 5, there was no correlation between quadriceps and hamstring for any index of neuromuscular fatigue (MVC,  $P_{TW}$ , or VA) measured immediately after the 5-min all-out protocol (MVC,  $r^2 = 0.137$ ,  $P = 0.238$ ;  $P_{TW}$ ,  $r^2 = 0.000$ ,  $P = 0.990$ ; VA,  $r^2 = 0.065$ ,  $P = 0.425$ ).

**Parameters associated with the torque–duration relationship.** As illustrated in Figure 6, critical torque was  $\sim 1.6$  times lower in hamstring compared with quadriceps when expressed in absolute units ( $60 \pm 12$  vs  $97 \pm 26$  N·m,  $P < 0.001$ ).



**FIGURE 4**—Neuromuscular fatigue during the 5-min all-out exercise protocol and recovery kinetics. MVC (A),  $P_{TW}$  (B), VA (C), and  $D_{10}/D_{100}$  ratio (D) assessed during exercise (gray areas) and recovery (white areas).  $D_{10}/D_{100}$ , low-frequency fatigue ratio. Data are presented as means  $\pm$  SD. Significant time, condition, and interaction effects are indicated on the graphs for both exercise (gray boxes) and recovery (white boxes).

TABLE 1. Effects of 5-min all-out exercise on neuromuscular variables measured immediately at exercise termination and during recovery.

| Parameters                  | Condition  | Pre         | Post        | Recovery   |            |             |             |             |            |
|-----------------------------|------------|-------------|-------------|------------|------------|-------------|-------------|-------------|------------|
|                             |            |             |             | 1 Min      | 2 Min      | 3 Min       | 5 Min       | 10 Min      | 15 Min     |
| D <sub>10</sub> (N·m)       | Quadriceps | 115 ± 25    | 34 ± 17*    | 56 ± 20*   | 65 ± 17*a  | 67 ± 13*    | 64 ± 11*    | 59 ± 13*    | 57 ± 15*   |
|                             | Hamstring  | 18 ± 6**    | 8 ± 3**     | 8 ± 3**    | 8 ± 3**    | 8 ± 3**     | 8 ± 3**     | 8 ± 3**     | 8 ± 3**    |
| D <sub>100</sub> (N·m)      | Quadriceps | 105 ± 19    | 54 ± 18*    | 65 ± 18*   | 71 ± 16*   | 73 ± 15*    | 74 ± 14*    | 73 ± 15*    | 72 ± 15*   |
|                             | Hamstring  | 22 ± 8**    | 13 ± 5**    | 13 ± 5**   | 14 ± 6**   | 14 ± 6**    | 14 ± 6**    | 14 ± 5**    | 14 ± 6**   |
| MRTD (N·m·s <sup>-1</sup> ) | Quadriceps | 1612 ± 309  | 564 ± 281*  | 838 ± 329* | 973 ± 327* | 1035 ± 274* | 1016 ± 233* | 1008 ± 230* | 987 ± 245* |
|                             | Hamstring  | 432 ± 160** | 245 ± 90*   | 257 ± 91** | 251 ± 82** | 245 ± 88**  | 244 ± 86**  | 246 ± 88**  | 235 ± 87** |
| CT (ms)                     | Quadriceps | 118 ± 14    | 112 ± 25    | 120 ± 20   | 122 ± 20   | 118 ± 20    | 114 ± 18    | 104 ± 10*   | 102 ± 8*   |
|                             | Hamstring  | 95 ± 10**   | 99 ± 100    | 101 ± 11   | 100 ± 11   | 99 ± 11     | 99 ± 11     | 98 ± 10     | 98 ± 10    |
| HRT (ms)                    | Quadriceps | 59 ± 13     | 90 ± 23*    | 88 ± 17*   | 82 ± 18*   | 80 ± 14*    | 74 ± 11*    | 65 ± 12     | 59 ± 11    |
|                             | Hamstring  | 42 ± 23     | 43 ± 18**   | 43 ± 16**  | 44 ± 17**  | 42 ± 15**   | 40 ± 13**   | 36 ± 10**   | 33 ± 8**   |
| M-waves amplitudes (mV)     | VL         | 15.8 ± 3.0  | 15.5 ± 3.4  | 15.8 ± 4.0 | 15.7 ± 3.8 | 15.5 ± 3.8  | 15.3 ± 3.9  | 14.7 ± 3.9  | 14.4 ± 3.8 |
|                             | VM         | 20.7 ± 5.8  | 20.2 ± 5.9  | 20.7 ± 5.7 | 20.8 ± 5.6 | 20.7 ± 5.5  | 20.6 ± 5.6  | 19.9 ± 5.7  | 19.6 ± 5.6 |
|                             | RF         | 9.3 ± 3.9   | 8.5 ± 3.5   | 9.3 ± 3.6  | 9.5 ± 3.5  | 9.4 ± 3.5   | 9.2 ± 3.3   | 8.7 ± 3.0   | 8.5 ± 3.0  |
|                             | BF         | 9.7 ± 4.8   | 12.8 ± 10.6 | 10.1 ± 5.0 | 10.1 ± 5.0 | 9.8 ± 4.9   | 9.4 ± 4.9   | 8.7 ± 4.4   | 8.7 ± 4.4  |
|                             | ST         | 8.6 ± 3.0   | 7.9 ± 2.9   | 8.3 ± 3.7  | 8.4 ± 3.8  | 8.2 ± 3.6   | 7.8 ± 3.3   | 7.6 ± 3.4   | 7.5 ± 3.2  |

Data are expressed as means ± SD. MRTD, CT, and HRT were calculated from single P<sub>TW</sub>.

\*Significantly different from preexercise ( $P < 0.05$ ).

\*\*Significantly different from quadriceps ( $P < 0.05$ ).

CT, contraction time; D<sub>10</sub>, 10 Hz doublets; D<sub>100</sub>, 100 Hz doublets; HRT, half relaxation time; RF, rectus femoris; ST, semitendinosus; VL, vastus lateralis; VM, vastus medialis.

Contrastingly, when expressed in relative units (%MVC), critical torque was greater in hamstring than quadriceps ( $50\% \pm 7\%$  MVC vs  $33\% \pm 9\%$  MVC,  $P < 0.001$ ).  $\dot{W}$  was ~8 times lower in hamstring compared with quadriceps ( $1001 \pm 696$  vs  $8111 \pm 2073$  N·m·s, respectively,  $P < 0.001$ ).

## DISCUSSION

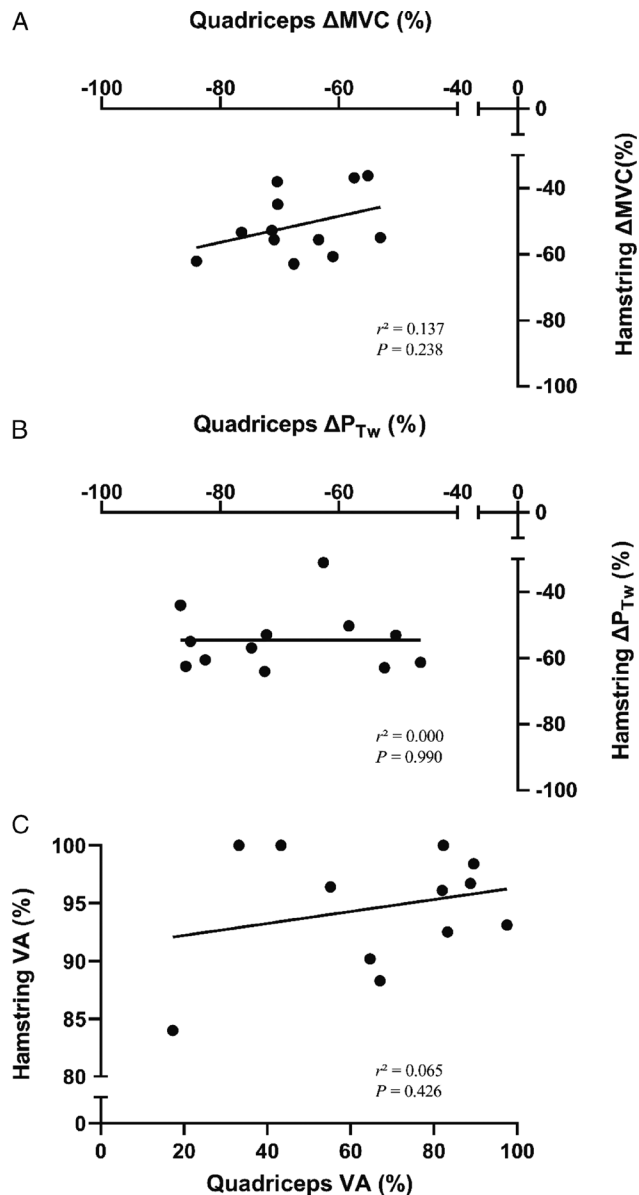
The present study investigated the mechanisms of neuromuscular fatigue in hamstring versus quadriceps muscles induced by an isometric exercise and its consequences on the torque–duration relationship in men. After an all-out exercise used to match exercise duration and relative intensity between muscle groups, hamstring appeared ~15% less fatigable ( $\Delta$ MVC) compared with quadriceps. Mechanistically, this discrepancy between muscle groups was driven by the development of less peripheral fatigue, and no central fatigue, compared with quadriceps. Additionally, exercise capacity was lower in hamstring compared with quadriceps, as a result of a drastically lower  $\dot{W}$  associated with a lower absolute critical torque. Thereafter, the recovery kinetics of peripheral fatigue was slower in hamstring compared with quadriceps. Importantly, there was no correlation between hamstring and quadriceps for any index of neuromuscular fatigue, indicating no interrelation between these two muscle groups.

**Neuromuscular fatigue in hamstring versus quadriceps muscles.** Peripheral fatigue was significantly lower in hamstring compared with quadriceps throughout exercise (Fig. 4), suggesting less intramuscular metabolic perturbation (22,23). As M-waves were unchanged in both hamstring and quadriceps (Table 1), peripheral fatigue was not explained by a reduction in membrane excitability and originated beyond the sarcolemma (24). Specifically, the difference observed in the D<sub>10</sub>/D<sub>100</sub> ratio, showing less low-frequency fatigue in hamstring compared with quadriceps, indicates that the attenuated peripheral fatigue observed in hamstring was attributed to a relatively preserved decrease in myofibrillar sensitivity to Ca<sup>2+</sup> and/or in the reduction of Ca<sup>2+</sup> released from the sarcoplasmic reticulum compared with quadriceps (25,26). This difference between the

two muscle groups might be explained, at least in part, by the greater proportion of fatigue-resistant Type I muscle fibers in hamstring compared with quadriceps (6,7). Of note, this result contrasts with a previous investigation showing that peripheral fatigue was greater in hamstring compared with quadriceps (8). Given the influence of the exercise duration/intensity on neuromuscular fatigue etiology (9), the discrepancy between studies might be explained by the drastically different exercise duration between muscle groups in Coratella et al. (8) while it was matched in the present investigation. Additionally, the selection of the muscle length (60° for both muscles in Coratella et al. vs the optimal angle of each muscle group in the present study) likely contributed to the differences observed in neuromuscular fatigue (27,28).

VA remained unaltered throughout exercise in hamstring, whereas a significant decrease was observed in quadriceps. If the absence of central fatigue in hamstring contrasts with what we and others found for this type of exercise in quadriceps (10,29), this result is supported by previous studies showing no decrease in VA after fatiguing hamstring exercises such as simulated soccer game (19) or repeated sprints (30). Mechanistically, the absence of central fatigue in hamstring might be related to group III/IV sensory neurons, which project from skeletal muscles to the central nervous system. Indeed, group III/IV muscle afferents feedback promotes central fatigue during exercise to restrict peripheral fatigue and to protect the exercising muscles from severe threats to homeostasis (31). Interestingly, peripheral fatigue, which is lower in hamstring compared with quadriceps, was identified to promote group III/IV muscle afferents firing (31,32). Therefore, group III/IV muscle afferents were likely more stimulated in quadriceps, leading to a significant decrease in VA during exercise only for this muscle group.

The substantial differences observed in the magnitude and etiology of neuromuscular fatigue between muscle groups might be related to the lower torque capacity of hamstring compared with quadriceps. Indeed, the classic correlation between maximal torque and fatigability (33) was also found in the present study (Fig. 3). Interestingly, the nature of this

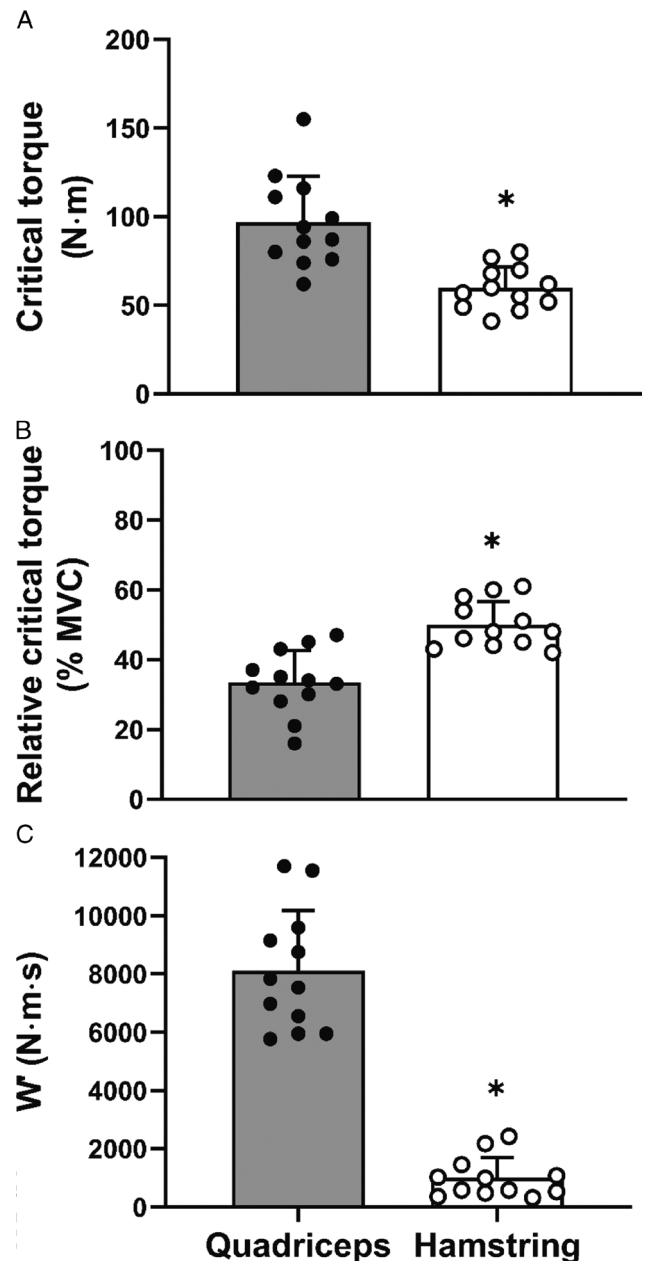


**FIGURE 5**—Correlations between quadriceps and hamstring for neuromuscular fatigue parameters. Correlation for MVC (A), peripheral fatigue (P<sub>TW</sub>, B), and central fatigue (VA, C) for quadriceps and hamstring. Data are expressed in percentage of decrease from baseline.

relationship was independent of the muscle group, suggesting that maximal torque might explain, at least in part, the higher development of neuromuscular fatigue observed in strong quadriceps compared with weaker hamstring muscles. Moreover, it was established that hamstring displayed a greater proportion in Type I muscle fibers than quadriceps (6,7), as evidenced indirectly in this study by a lower MRTD in hamstring (34,35). Given the influence of muscle typology on both fatigue resistance and muscle torque capacity (36), muscle typology is likely a major characteristic explaining the discrepancies we observed between hamstring and quadriceps muscles.

Exercise performance was lower in hamstring compared with quadriceps, as a result of a drastically lower  $W'$  associated with a lower absolute critical torque. The lower  $W'$  in hamstring

is consistent with the lower level of peripheral fatigue obtained following exercise using this muscle group, as these two parameters have been previously related (14,17). Moreover, critical torque has been related to skeletal muscle typology. Indeed, for a given muscle mass, a greater proportion in Type I muscle fibers was associated with a greater critical torque (13). Therefore, the greater proportion in Type I muscle fibers in hamstring (47% [6]) compared with quadriceps (34% [7]), indirectly evidenced in the present study by the lower MRTD in hamstring (34), likely explain why hamstring maintained a higher relative critical torque than quadriceps. However, absolute critical torque



**FIGURE 6**—Parameters of torque-duration relationship. Critical torque was expressed in newton-meter (A) and normalized as a percent of MVC (B). The work performed above critical torque ( $W'$ ) was expressed in newton-meters-second (C) for both quadriceps and hamstring muscles. \*Significant difference vs quadriceps ( $P < 0.05$ ).

was lower in hamstring, as muscle mass, which has a direct influence on the muscle torque capacity (37), was greater in quadriceps. Together, these results highlight a greater fatigue resistance and a reduced exercise capacity in hamstring compared with quadriceps in men.

**Recovery kinetics in hamstring versus quadriceps muscles.** Although hamstring and quadriceps demonstrated different magnitude and etiology in neuromuscular fatigue, recovery kinetics were also significantly different for all indices (Fig. 3). Indeed, although quadriceps was the only muscle group to display central fatigue, VA fully recovered within 2 min postexercise, which is consistent with previous results (38,39). Moreover, with a greater level of peripheral fatigue developed throughout exercise, quadriceps also recovered faster than hamstring. Regardless, both muscle groups followed a biphasic recovery kinetics for MVC and  $P_{TW}$  (40), with 1) a large recovery during the first minute postexercise, succeeded by 2) a slower recovery phase where MVC and  $P_{TW}$  remained depressed compared with baseline. Mechanistically, this prolonged low-frequency force depression (24), as evidenced in the present study by the  $D_{10}/D_{100}$  ratio (Fig. 4), is the consequence of a reduction in  $Ca^{2+}$  release into the sarcoplasmic reticulum due to intrinsic alterations in the  $Ca^{2+}$  pump (25,41).

**Is neuromuscular fatigue interrelated between hamstring and quadriceps muscles?** The present study documented the absence of interrelation between hamstring and quadriceps for all neuromuscular fatigue indices (Fig. 5), indicating that a highly fatigable individual in quadriceps is not necessarily highly fatigable in hamstring. Therefore, the fatigability that an individual can develop in quadriceps cannot be used to extrapolate his/her fatigability in hamstring. Although muscle groups presented very different fatigue etiology, the absence of interrelation between quadriceps and hamstring was not surprising. Moreover, the interrelation in peripheral fatigue between muscles might be limited to those with similar fiber type proportion (42), which is not the case for quadriceps and hamstring (6,7). For central fatigue, no correlation was found between hamstring and quadriceps, which is consistent with a previous study comparing fatigue in plantar flexors and finger flexors (43). In conclusion, although neuromuscular fatigue research focusing on hamstring is scarce compared with quadriceps or calf muscles (44), there is a need to investigate specifically hamstring muscles fatigue.

**Perspectives for hamstring muscle injuries.** Given the higher injury prevalence in hamstring than quadriceps in sports requiring sprints and high-intensity actions (3,45,46) and the identification of fatigue as a risk factor for injuries (2,47–50), observation of less neuromuscular fatigue in hamstring versus quadriceps might seem paradoxical. However, although lower, the magnitude of peripheral fatigue observed in hamstring was still substantial (~55%), which cannot rule out the implication of fatigue in hamstrings injuries. Moreover, preexercise MVC and absolute critical torque were both lower in hamstring compared with quadriceps. Importantly, a deficit in maximal strength, leading the muscle to operate closer to its mechanical limit, has been identified as a risk factor for hamstring

muscle injuries (51,52). Therefore, either or both muscle strength and fatigue, two interconnected parameters (33,53), are likely involved in hamstring muscle injuries. Further studies are needed to dissociate and quantify their relative contribution.

In practice, although greater preexercise torque is associated with greater fatigability (Fig. 3), one could argue that specific resistance training might be counterproductive for injury prevention by decreasing critical torque. However, the present study showed that the relationship between torque output and fatigue was not parallel to the identity line (i.e., less steep, see Fig. 3), indicating that for a given amount of force gained, the amount of fatigue exacerbated is relatively smaller (by 21%). In other words, the extra MVC torque earned is greater than the associated decrease in critical torque, and resistance training will lead to an increase in absolute critical torque.

The exercise modality selected in the present study was designed to activate independently quadriceps or hamstring muscles during exercise; however, it did not reproduce specifically the physiological demand observed during whole-body exercise “on the field.” For example, the number of muscle groups involved during exercise influences the magnitude of neuromuscular fatigue and exercise performance (31,54). Further studies are needed to characterize neuromuscular fatigue in hamstring and quadriceps during competitions and typical training sessions with a high risk of hamstrings muscle injuries such as sprints or agility training. This will help to delineate the role of exercise-induced fatigue in the occurrence of injuries, documented as a multifactorial event (55).

This study was only performed on men, and given existing sex differences in skeletal muscle typology (56), maximal force, and neuromuscular fatigue development (57,58), the conclusion may not be generalizable to women.

## CONCLUSIONS

The present study documented differences in neuromuscular fatigue etiology between quadriceps and hamstring muscles while exercise was matched for duration and relative intensity in men. Hamstring developed less peripheral fatigue, and no central fatigue, during the 5-min all-out protocol. Moreover, exercise performance was lower in hamstring compared with quadriceps, as a result of a drastically lower  $W'$  associated with a lower absolute critical torque. The absence of correlation observed between quadriceps and hamstring fatigue parameters suggests no interrelation in fatigue etiology between both muscle groups within individuals and, therefore, highlights the need to investigate specifically hamstring muscle fatigue (e.g., to better understand hamstring muscle injuries).

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